The Canadian Urban Environmental Health Research Consortium

Canue Metadata - Weather Land Surface Temperature v1

2026-11-06

DATA SET INFORMATION

Dataset Code: WTLST_AVA_YY

Description:

CANUE staff developed annual estimates of maximum mean warm-season land surface temperature (LST) recorded by LandSat 8 at 30m resolution. To reduce the effect of missing data/cloud cover/shadows, the highest mean warm-season value reported over three years was retained - for example, the data for 2021 represent the maximum of the mean land surface temperature at a pixel location between April 1st and September 30th in 2019, 2020 and 2021. Land surface temperature was calculated in Google Earth Engine, using a public algorithm (see supplementary documentation). In general, annual mean LST may not reflect ambient air temperatures experienced by individuals at any given time, but does identify areas that are hotter during the day and therefore more likely to radiate excess heat at night - both factors that contribute to heat islands within urban areas.

Keywords: heat islands|temperature

Place Keywords: national|Canada

GEOSPATIAL REFERENCE

Upper Left Corner: 65.14N , -141.02W

Lower Right Corner: 41.68N , -52.62W

Coordinate System: GCS_WGS84 - EPSG:4326

Geometry Type: POINT - Units: Decimal Degree

Geometry Data Source: DMTI Spatial Inc. (postal codes)

QUALITY ASSESSMENT

QA/QC Procedures:

CANUE staff did not assess the quality of LandSat data or the accuracy of the LST algorithm vs ground based temperature monitors.

Geographic Coordinate Positional Accuracy:

These metrics are linked to the corresponding annual postal codes files for mapping and analysis purposes. Refer to the postal code metadata file in Supporting Documentation for more information.

Vertical Positional Accuracy:

Attribute Accuracy:

Data Validity: NoData = -9999 (for numeric fields) - NoData=null (for category fields) - Data insufficient to calculate value = -1111

Associated Files:

Data Comment:

DATA SOURCE

Data Source

LandSat 8 via Google Earth Engine and LST algorithm (see support documentation)

Spatial Resolution: 30 metres

Data Preparation Date: 2022-03-01

Beginning Date: 2015

End Date: 2021

Sampling Frequency of Data: Annual

Years Available:
2015 - 2016 - 2017 - 2018 - 2019 - 2020 - 2021

MAINTENANCE

Description:

File Type: Comma separated values(.csv)

File Size:

Number of Data Files: 7

DATA USE CONDITIONS

The Data User is REQUIRED:
(i) to acknowledge data sources listed under Acknowledgement(s)
(ii) cite the publication(s) listed under Recommended Citation(s) as the providers and source of these data when using them in support of research, analysis, operations, policy decision or any other undertaking including publication
(iii) complete and sign the CANUE Data Use and Sharing Agreement in which the name and signature of the researcher/analyst who takes responsibility for ensuring all conditions are met.

Data Sharing Restrictions:
These data files are provided solely for the purposes stated in the CANUE Data Sharing and Use Agreement and should not be re-distributed for any reason. These data also contain proprietary postal code data and may only be used for the project named in the CANUE Data Sharing and Use Agreement. Data can be shared only within a project team for the exclusive purposes of teaching, academic research and publishing, and/or planning of educational services in accordance to DMTI End User Agreement associated with the Spatial Mapping Academic Research Tools (SMART) Program.

Include the following references in any publications resulting from the use of these data:
[1] Setton, E; Redivo A. (2022). Three-year warm-season maximum land surface temperature from LandSat 8. Developed for the Canadian Urban Environmental Health Research Consortium. [2] Osei, Edwin (2022). Google Earth Engine code for calculating LST using LandSat 8. Computer code, posted on Stack Exchange April 22, 2022.
[3] CanMap Postal Code Suite [compute file] DMTI Spatial Inc,. various years.

Include the following acknowledgements:
Metrics of land surface temperature, indexed to DMTI Spatial Inc. postal codes were provided by CANUE (Canadian Urban Environmental Health Research Consortium).

SUPPORT DOCUMENTATION

1 - LandSat 8 Surface Reflectance Tier 1 - Google Earth Engine (https://developers.google.com/earth-engine/datasets/catalog/LANDSAT_LC08_C01_T1_SR)

2 - Google Earth Engine land surface temperature code (https://canuedata.ca/docs/LST_GEE-code.docx)

3 - Postal code metadata (https://canuedata.ca/docs/CANUE-Browser-Metadata-PostalCodes.pdf)

VARIABLES

WTLSTYY_01 - 3 Years Annual Max of Means at Postal Code

Highest mean warm-season temperature in last three years at postal code location in Celsius

WTLSTYY_02 - 3 Years Annual Max of Means 100m

Average of the highest mean warm-season temperature in last three years across all pixels within 100m of postal code location in Celsius

WTLSTYY_03 - 3 Years Annual Max of Means 250m

Average of the highest mean warm-season temperature in last three years across all pixels within 100m of postal code location in Celsius

WTLSTYY_04 - 3 Years Annual Max of Means 500m

Average of the highest mean warm-season temperature in last three years across all pixels within 500m of postal code location in Celsius

WTLSTYY_05 - 3 Years Annual Max of Means 750m

Average of the highest mean warm-season temperature in last three years across all pixels within 750m of postal code location in Celsius

WTLSTYY_06 - 3 Years Annual Max of Means 1Km

Average of the highest mean warm-season temperature in last three years across all pixels within 1000m of postal code location in Celsius

SUPPORT CONTACT

Data Set Support Contact: info@canue.ca

Affiliated Organization:
CANUE (Canadian Urban Environmental Health Research Consortium)
Dalla Lana School of Public Health, University of Toronto

WebSite: https://www.canue.ca

Toronto - Ontario - Canada

DATA SOURCE CONTACT

Data Set Support Contact:

Email:

Affiliated Organization:

- -